

Experience

- Graduate Machine Learning Engineer — Graphcore, Cambridge** 2026–
- Large-scale distributed LLM pre-training across multi-rack GPU clusters.
- ML Research Engineer (Contract) — Ascentra Labs, London** 2025
- Designed and delivered proof-of-concept for LLM integration into survey analysis product.
 - Built pipeline converting natural language queries to JSON requests.
- Postgraduate Researcher — University College London** 2024–2026
- Research on Neural Program Synthesis and Algorithmic Reasoning in the Machine Intelligence Lab, supervised by Prof. Mirco Musolesi. Developed a transformer-based neural program synthesis framework for domain-agnostic library learning.
- Research Intern — Imperial College London** 2023
- Causal discovery using cross-validation as compared with Bayesian Model Selection (using GPLVM), supervised by Mark van der Wilk and funded by Imperial UROP. Implemented models and ran large-scale experiments on Imperial’s HPC cluster.

Education

- University College London** September 2023 – September 2024
MSc Machine Learning (*Distinction*)
- **MSc Thesis** (*Distinction*): *Hierarchical Bayesian Program Synthesis for Neural Algorithmic Reasoning*, supervised by Prof. Mirco Musolesi.
 - Completed PhD-level modules at the Gatsby Computational Neuroscience Unit.
- Imperial College London** October 2020 – June 2023
BSc (Honours) Physics with Theoretical Physics (*2:1*)
- **BSc Thesis** (*First-class*): *Computational Evolution*, supervised by Dr David Clements.

Projects

- Transformer Program Synthesis — University College London** 2025
- Developed novel programming-by-example framework: Transformer-based model to recover algorithms from data, training neural functions on-the-fly.
- Masked Image Modelling — University College London** 2024
- Self-supervised learning framework: VisionTransformer pre-trained to reconstruct partially masked images and fine-tuned for image segmentation. (*Distinction*)
- Machine learning for particle physics — Imperial College London** 2022
- Lead ML team analysing decay product distribution from a Large Hadron Collider experiment (LHCb) for Standard Model comparison. TensorFlow, XGBoost. (*First-class, highest score*)

Skills & Languages

Python (PyTorch, TensorFlow, GPflow, JAX), Julia; GPU clusters, distributed training; Git, Linux
English, Hungarian, French (Intermediate)